

HDOC Day-12

- Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations:
- Find the sum and product of the first and last digits. E.g- Number: 12345, Sum= 6 , Product=5
- Count how many digits are lesser than, equal , and greater than a given digit 'k'. E.g- Number:12345, k=3, Count_lesser=2, Count_equal=1, Count_greater=2

- **1. Algorithm:**

Step 1: Start

Step 2: Input a positive integer n.

Step 3: Find the last digit using $n \% 10$.

Step 4: Find the first digit by repeatedly dividing n by 10 until $n < 10$.

Step 5: Calculate:

sum = first_digit + last_digit

product = first_digit $\times$ last_digit

Step 6: Display the sum and product.

Step 7: Stop

- **Example:**

- For 12345:
- First digit = 1
- Last digit = 5
- Sum = $1 + 5 = 6$
- Product = $1 \times 5 = 5$

- **2. Algorithm**

Step 1: Start

Step 2: Input a positive integer n

Step 3: Input the digit k

Step 4: Initialize:

count_less = 0, count_equal = 0, count_greater = 0

Step 5: Extract each digit using $n \% 10$.

Step 6: Compare the digit with k:

If digit < k, increment count_less

If digit == k, increment count_equal

If digit > k, increment count_greater

Step 7: Remove the last digit using $n = n / 10$

Step 8: Repeat until all digits are checked

Step 9: Display all three counts

Step 10: Stop

● **Example:**

- For 12345, k = 3:
- Less than 3 → 1, 2 → 2
- Equal to 3 → 3 → 1
- Greater than 3 → 4, 5 → 2

● **C Program:**

```
#include <stdio.h>
int main()
{
    int choice, n, k;
    int first, last, sum, product;
    int digit, count_less, count_equal, count_greater;

    printf("Enter a positive integer: ");
```

```

scanf("%d", &n);

printf("\n----- MENU ----- \n");
printf("1. Find sum and product of first and last digits\n");
printf("2. Count digits less than, equal to and greater than k\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        /* Find last digit */
        last = n % 10;

        /* Find first digit */
        first = n;

        while(first >= 10)
        {
            first = first / 10;
        }

        sum = first + last;
        product = first * last;

        printf("\nFirst digit = %d", first);
        printf("\nLast digit = %d", last);
        printf("\nSum = %d", sum);
        printf("\nProduct = %d", product);

        break;

    case 2:
        printf("Enter the digit k (0-9): ");
        scanf("%d", &k);

        count_less = 0;
        count_equal = 0;
        count_greater = 0;

        while(n > 0)

```

```

{
    digit = n % 10;

    if(digit < k)
        count_less++;
    else if(digit == k)
        count_equal++;
    else
        count_greater++;

    n = n / 10;
}

printf("\nCount of digits less than %d = %d", k, count_less);
printf("\nCount of digits equal to %d = %d", k, count_equal);
printf("\nCount of digits greater than %d = %d", k, count_greater);

break;

default:
    printf("\nInvalid choice!");

}

return 0;
}

```

• Test Case Table: For Choice 1:

Test Case	Input	Expected First Digit	Expected Last Digit	Sum	Product
1	12345	1	5	6	5
2	9876	9	6	15	54
3	1001	1	1	2	1
4	56789	5	9	14	45

For Choice 2:

Test Case	Input	k	Less than	Equal	Greater than
1	12345	3	2	1	2
2	5555	5	0	4	0
3	2468	5	2	0	2
4	11123	1	0	3	2

Compilation and Execution:

The following test cases have been compiled and executed successfully.

Choice 1:

Test Case 1:

```
(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:12345

—— MENU ——
1. Find sum and product of first and last digits
2. Count digits less than,equal to and greater than k
Enter your choice: 1

First digit = 1
Last digit = 5
Sum = 6
Product = 5
```

Test Case 2:

```
(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:9876

—— MENU ——
1. Find sum and product of first and last digits
2. Count digits less than,equal to and greater than k
Enter your choice: 1

First digit = 9
Last digit = 6
Sum = 15
Product = 54
```

Test Case 3:

```
(kali@kali)-[~]  
$ ./a.out  
Enter a positive integer:1001  
  
—— MENU ——  
1. Find sum and product of first and last digits  
2. Count digits less than,equal to and greater than k  
Enter your choice: 1  
  
First digit = 1  
Last digit = 1  
Sum = 2  
Product = 1
```

Test Case 4:

```
(kali@kali)-[~]  
$ ./a.out  
Enter a positive integer:56789  
  
—— MENU ——  
1. Find sum and product of first and last digits  
2. Count digits less than,equal to and greater than k  
Enter your choice: 1  
  
First digit = 5  
Last digit = 9  
Sum = 14  
Product = 45
```

Choice 2:

Test Case 1:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer:12345  
  
—— MENU ——  
1. Find sum and product of first and last digits  
2. Count digits less than,equal to and greater than k  
Enter your choice: 2  
Enter the digit k (0-9): 3  
  
Count of digits less than 3 = 2  
Count of digits equal to 3 = 1  
Count of digits greater than 3 = 2
```

Test Case 2:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer:5555  
  
—— MENU ——  
1. Find sum and product of first and last digits  
2. Count digits less than,equal to and greater than k  
Enter your choice: 2  
Enter the digit k (0-9): 5  
  
Count of digits less than 5 = 0  
Count of digits equal to 5 = 4  
Count of digits greater than 5 = 0
```

Test Case 3:

```
(kali@kali)-[~]  
$ ./a.out  
Enter a positive integer:2468  
  
—— MENU ——  
1. Find sum and product of first and last digits  
2. Count digits less than,equal to and greater than k  
Enter your choice: 2  
Enter the digit k (0-9): 5  
  
Count of digits less than 5 = 2  
Count of digits equal to 5 = 0  
Count of digits greater than 5 = 2
```

Test Case 4:

```
(kali@kali)-[~]  
$ ./a.out  
Enter a positive integer:11123  
  
—— MENU ——  
1. Find sum and product of first and last digits  
2. Count digits less than,equal to and greater than k  
Enter your choice: 2  
Enter the digit k (0-9): 1  
  
Count of digits less than 1 = 0  
Count of digits equal to 1 = 3  
Count of digits greater than 1 = 2
```